Final Project: News

Dexter Paclebar

University of Advancing Technology

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Professor Rawad Habib

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Project Overview

Staying informed on world events often requires navigating multiple news sources, which can be time-consuming and overwhelming. Many people are interested in global news but do not know where to start looking. The *News* mobile application provides users with a simple, fast, and visually clean way to stay informed about current events around the globe. It fetches articles from at least one online news API and displays them in English, allowing users to filter and search by keywords without requiring user accounts or stored personal data. The News application gathers global articles from an online API, providing minimal yet relevant information: title, thumbnail, description, publication date/time, and source; along with filtering and keyword search capabilities. Users can select an article to view its original source in their mobile browser. Individuals who want to quickly see a variety of world news articles in English without committing to a single news outlet or creating an account will benefit most from this application.

Objectives

In-Scope

- Fetching articles from at least one news API.
- Displaying article title, thumbnail, description, publication date/time, and source.
- Searching articles by keyword in the title, description, or publishing date.
- Filtering displayed articles by user-selected keywords.
- Updating the news feed at least every 5 minutes.
- Refreshing the feed without requiring a full page reload.
- Light and dark mode display options.
- Swapping between APIs without major code changes.

Out-of-Scope

- Original journalism or user-created content.
- User accounts, personal bookmarking, or article saving.
- Offline reading mode.

- Additional commentary or opinion-based content.

Requirements

Functional

- The application will fetch news articles from at least one online news API.
- The application will display the title, thumbnail, description, publication date, publication time, and source for each article.
- The application will allow users to filter articles using user-defined keywords.
- The application will allow keyword searches across article titles, descriptions, and publishing dates.
- The application will update the news content at least every 5 minutes.
- The application will refresh displayed content without requiring a full page reload.
- The application will allow the user to switch between light and dark mode.

Non-Functional

- Performance: The application shall load the home page in under 5 seconds.
- Performance: The application shall update displayed news without requiring a full page reload.
- Usability: The interface will require no more than two taps to reach any news article.
- Privacy: The application shall not require user data beyond cache storage.
- Maintainability: The application will support API source changes without major code refactoring.
- Compatibility/Portability: The application shall support light and dark mode on mobile devices.

Use Case/UML Diagram

Use Case

Actors:

- User: Browses and searches articles.
- News API: Provides article data.
- System/Application: Fetches, filters, and displays articles.

Preconditions:

- The application is running and connected to at least one news API source.

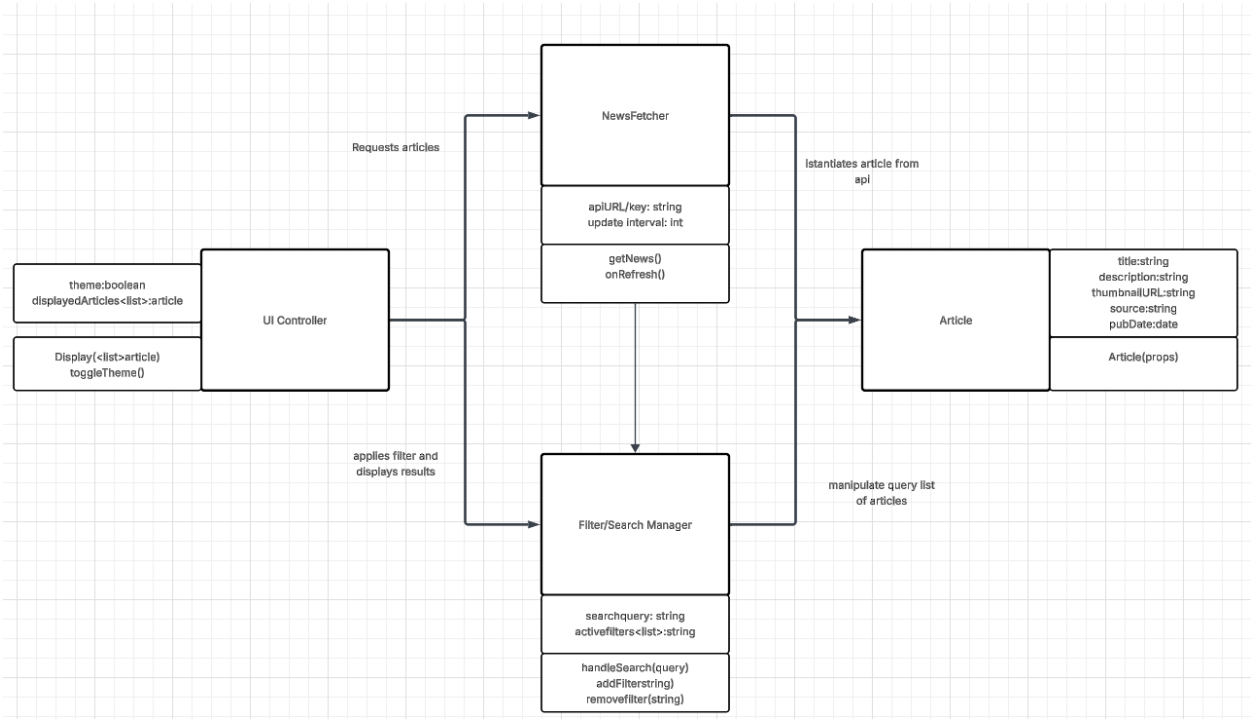
Postconditions:

- The user can see up-to-date news articles based on selected preferences.

Basic Flow (Main Success Scenario):

1. User opens the *News* application.
2. The system fetches articles from the API.
3. The system displays articles (filtered if applicable).
4. The user browses the articles.
5. The user clicks on an article and is redirected to the original source.

UML Diagram



Glossary

- API (Application Programming Interface): A protocol for interacting with software services, in this case for retrieving news data.
- Cache Data: Temporarily stored data used to speed up application performance.
- Thumbnail: A small image representing the article visually.

Assumptions

- The news API will provide JSON-formatted data containing article title, description, image URL, publication date/time, and source.
- The news API will remain available and operational during app use.
- Users will have an active internet connection.
- All articles fetched will be in English.